Voter Info App: Product Requirements Document

Version 1.0 · October 27, 2026 · Stack: static web app + Netlify Functions, no database

1. Summary

A single web app where a US voter enters their address once and gets everything they need for an upcoming election: where to vote, a personal sample ballot, plain-language explanations of ballot measures, their districts and officials, and reminders. Data comes from the Google Civic Information API (plus free open datasets for officials). The app helps people **prepare to vote**; it does not cast or record official votes.

2. Goals and non-goals

Goals

- One address entry powers every feature.
- Works on mobile first; loads fast on Netlify's CDN.
- No database and no user accounts. All personal data stays in the user's browser.
- API keys never reach the browser (all external calls go through Netlify Functions).

Non-goals (v1)

- Casting or submitting real ballots, or any voter registration submission.
- Candidate endorsements or recommendations. The app stays neutral.
- Coverage outside the US.

3. Users

- **First-time or infrequent voters** who don't know where to vote or what is on the ballot.
- **Non-native English speakers** who need ballot measures explained simply or in another language (e.g., Vietnamese, Spanish).
- **Engaged citizens and students** who want to know their districts and representatives, or explore polling-access data.

4. Features

The six capabilities below map to the app ideas in the companion Civic API PDF, combined into one product.

#	Feature	Requirements	Data source
F1	Where do I vote	Show polling place, early-vote sites and drop boxes for the address on a map; hours; one-tap directions link (Google Maps URL).	`/voterinfo` + Maps Embed/JS API

#	Feature	Requirements	Data source
F2	Personal sample ballot	List every contest and measure; candidate name, party, website/social links; user can mark picks, then print or download PDF. Picks saved in localStorage only.	`/voterinfo`
F3	Ballot measure explainer	For each referendum: plain-language summary, what a Yes/No vote means, optional translation. Neutral wording; always show original text alongside.	`/voterinfo` + LLM via Netlify Function
F4	Election reminders	List upcoming elections; user downloads .ics calendar events (early voting start, Election Day) and can enable browser notifications. No email/SMS in v1 (would need storage).	`/elections`, `/voterinfo`
F5	Know your districts & officials	Show congressional, state and local districts; list federal and state officials with contact links.	`divisionsByAddress` + congress-legislators data + OpenStates
F6	Polling access insights	Page with charts by county: voters per polling site, sites per area. Data precomputed at build time into static JSON.	`/voterinfo` (build script) + Census data

5. Core user flow

- Landing page: address field with autocomplete, plus a short note that the address is not stored on any server.
- Dashboard: cards for each feature (F1 to F6), with the next election date at the top.
- If no election data exists for the address (off-season), show districts/officials and a clear "no upcoming election data yet" message, and offer reminders.
- Address saved in localStorage so returning users skip step 1; a "Clear my data" button removes it.

6. Architecture (Netlify, no database)

Layer	Choice	Notes
Frontend	Static SPA (React + Vite, or plain HTML/JS)	Deployed to Netlify CDN; mobile-first.
API proxy	Netlify Functions	`/api/voterinfo`, `/api/elections`, `/api/districts`, `/api/officials`, `/api/explain`. Holds API keys in Netlify environment variables.
Caching	HTTP cache headers + Netlify CDN	Cache `/elections` ~1 hour; voterinfo ~15 min per address hash. Cuts API quota use.
User state	Browser localStorage	Saved address, ballot picks, language choice. Never sent to a server except the address for lookups.
Static data	JSON files built at deploy time	Federal legislators list and F6 county stats; refreshed by a scheduled Netlify build hook.

Layer	Choice	Notes
Explainer cache	Static JSON or in-memory per function	No DB: optionally pre-generate measure summaries at build time; otherwise generate on request.

7. Non-functional requirements

- **Privacy:** no accounts, no analytics that log addresses, no address in URLs or logs.
- **Accessibility:** WCAG 2.1 AA; screen-reader friendly ballot; large tap targets.
- **Performance:** first load under 2 s on 4G; lookups under 3 s.
- **Neutrality:** candidates shown in API order; LLM prompts require neutral, sourced summaries with a disclaimer.
- **Reliability:** friendly error states when the API returns no data or rate-limits.
- **Languages:** UI in English first; explainer translations for Spanish and Vietnamese in v1.

8. Risks and constraints

- Civic API voter data is seasonal and coverage varies by state; the app must handle empty results gracefully.
- Google removed the Representatives endpoint; officials data depends on third-party datasets that may lag.
- LLM summaries can be wrong; always show official text and link to the state election site.
- Without a database, email/SMS reminders and shared ballots are out of scope for v1.

9. Milestones

Phase	Scope	Done when
1. Foundation	Netlify setup, functions proxy, address entry, F1	Test election `2000` shows polling places on a map.
2. Ballot	F2 sample ballot + print/PDF, F3 explainer	Measures show summaries; picks persist after reload.
3. Civic info	F5 districts & officials, F4 reminders (.ics)	Calendar file imports correctly into Google/Apple Calendar.
4. Insights & polish	F6 analytics page, accessibility, translations	Lighthouse accessibility 95+; charts render from static JSON.

10. Success metrics

- Share of address lookups that return usable election data.
- Sample ballots printed or downloaded.

- Reminder .ics downloads per election.
- Errors per 1,000 lookups under 1%.